This means, for a given magnitude of $\vec{E}$- field and $\vec{B}$- field, the forces act only for the particle moving with particular speed $v_o = \frac{E}{B}$. This speed is independent of mass and charge.

If the charge enters into the crossed fields with velocity v, other than $v_o$, it results in any of the following possibilities (Table 3.4).

**Table 3.4 Deflection based on the velocity – velocity selector**

| S.No. | Velocity | Deflection |
|---|---|---|
| 1 | $v > v_o$ | Charged particle deflects in the direction of Lorentz force |
| 2 | $v < v_o$ | Charged particle deflects in the direction of Coulomb force |
| 3 | $v = v_o$ | No deflection and particle moves straight |

So by proper choice of electric and magnetic fields, the particle with particular speed can be selected. Such an arrangement of fields is called a **velocity selector**.

**Note**

This principle is used in Bainbridge mass spectrograph to separate the isotopes.

## EXAMPLE 3.25

Let E be the electric field of magnitude $6.0 \times 10^6$ N $C^{-1}$ and B be the magnetic field magnitude 0.83 T. Suppose an electron is accelerated with a potential of 200 V, will it show zero deflection?. If not, at what potential will it show zero deflection.

***Solution*:**

Electric field, E = $6.0 \times 10^6$ N $C^{-1}$ and magnetic field, B = 0.83 T.

Then

$$v = \frac{E}{B} = \frac{6.0 \times 10^6}{0.83} = 7.23 \times 10^6 \, m s^{-1}$$

When an electron goes with this velocity, it shows null deflection. Since the accelerating potential is 200 V, the electron acquires kinetic energy because of this accelerating potential. Hence,

$$\frac{1}{2} m v^2 = eV \Rightarrow v = \sqrt{\frac{eV}{2m}}$$

Since the mass of the electron, $m = 9.1 \times 10^{-31} \, kg$ and charge of an electron, $|q| = e = 1.6 \times 10^{-19} C$. The velocity due to accelerating potential 200 V

$$v_{200} = \sqrt{\frac{2\left(1.6 \times 10^{-19}\right)(200)}{\left(9.1 \times 10^{-31}\right)}} = 8.39 \times 10^6 \, m \, s^{-1}$$

Since the speed $v_{200}$ > v, the electron is deflected towards direction of Lorentz force. So, in order to have null deflection, the potential, we have to supply is

$$V = \frac{1}{2} \frac{m v^2}{e} = \frac{\left(9.1 \times 10^{-31}\right) \times \left(7.23 \times 10^6\right)^2}{2 \times \left(1.6 \times 10^{-19}\right)}$$

$$V = 148.65 \, V$$

## 3.10.4 Cyclotron

Cyclotron (Figure 3.54) is a device used to accelerate the charged particles to gain large kinetic energy. It is also called as high energy accelerator. It was invented by Lawrence and Livingston in 1934.

**Principle**

When a charged particle moves normal to the magnetic field, it experiences magnetic Lorentz force.



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